
Cell Segmentation in Histopathology Images: Architecture Comparison and Ablation Study

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1 Overview

Pathologists manually analyze tissue samples to identify and classify abnormal cells—a process that is time-consuming, subjective, and prone to inter-observer variability. Automated cell segmentation can accelerate diagnostic workflows and improve reproducibility. This project builds a deep learning pipeline that segments cell nuclei in histopathology image patches and systematically compares architectural choices—spanning CNNs and Transformers—through controlled ablation experiments.

The project proceeds in two phases. First, a baseline MLP patch classifier is implemented from scratch (dense layers, ReLU, cross-entropy, SGD with momentum) to establish the data pipeline and a performance floor, directly exercising concepts from Lectures 3–5. Second, a U-Net encoder-decoder architecture Ronneberger et al. [2015] is trained for pixel-wise nuclei segmentation, followed by a series of ablation studies that each isolate a single architectural variable:

Experiment	Variation A	Variation B	Variation C	Course Connection
Encoder backbone	VGG-style 3×3 convs	ResNet-style residuals	Swin Transformer	Lec 08 / Lec 11: Architectures
Batch normalization	Conv→ReLU	Conv→BN→ReLU	—	Lec 08: BatchNorm
Loss function	Cross-entropy	Dice loss	—	Lec 03: Loss design
Network depth	3 downsample stages	4 downsample stages	—	Lec 07: Receptive fields
Initialization	Xavier	MSRA/He	—	Lec 05: Weight init

Each experiment is documented with training curves, final metrics (Dice > 0.75 , IoU > 0.65), and discussion of *why* results differ—connecting back to gradient flow, internal covariate shift, receptive field analysis, and the role of self-attention versus local convolutions. The final deliverable includes a qualitative failure analysis with predicted masks overlaid on input images, and a reproducible inference pipeline.

2 Related Work

U-Net Ronneberger et al. [2015] introduced the symmetric encoder-decoder architecture with skip connections that concatenate high-resolution encoder features to the decoder path, enabling precise localization. Originally demonstrated on biomedical images with limited training data, it remains the dominant segmentation backbone in medical imaging. Our project uses U-Net as the base architecture and studies how encoder design choices affect its performance.

Residual Networks He et al. [2016] showed that identity shortcut connections enable training of substantially deeper networks by addressing the degradation problem, where deeper plain networks paradoxically exhibit higher training error. He et al. [2015] separately proposed MSRA initialization (weights drawn from $\mathcal{N}(0, \sqrt{2/n_{in}})$), which is standard for ReLU networks. Our ablation study directly compares VGG-style and ResNet-style encoders within the U-Net framework, and tests MSRA versus Xavier initialization.

Vision Transformers Dosovitskiy et al. [2021] applied the Transformer architecture to image recognition by splitting images into fixed-size patches, embedding them linearly, and processing the sequence with standard self-attention layers. While ViTs require large-scale pretraining, hierarchical variants such as the Swin Transformer Liu et al. [2021] introduced shifted windows to compute self-attention locally, reducing cost to linear complexity and producing multi-scale feature maps compatible with dense prediction tasks. We include a Swin-based encoder in our U-Net comparison to evaluate whether Transformer-derived features improve nuclei segmentation on small (256×256) medical image patches relative to CNN encoders.

Batch Normalization Ioffe and Szegedy [2015] normalizes layer inputs to reduce internal covariate shift, enabling higher learning rates and faster convergence. It is now standard in most deep architectures but its interaction with residual connections and different loss functions remains empirically dataset-dependent. One of our ablations isolates the effect of BN in the segmentation setting.

PanNuke Gamper et al. [2019] provides a large-scale, multi-tissue benchmark for nuclei segmentation across 19 tissue types, pre-split into three folds. Gamper et al. [2020] extended the dataset to 189,744 annotated nuclei and demonstrated significant performance variation across tissue types, motivating architectural comparison studies such as ours.

HoVer-Net Graham et al. [2019] extended nuclei segmentation beyond binary masks by predicting horizontal and vertical distance maps to separate overlapping nuclei, achieving state-of-the-art results on several benchmarks including Kumar and CoNSeP. HoVer-Net was later applied to PanNuke in the dataset extension study Gamper et al. [2020]. While we do not replicate their distance-map approach, HoVer-Net’s results serve as an upper-bound reference for our experiments.

3 Dataset

We use **PanNuke** Gamper et al. [2019, 2020]: 7,904 images (256×256 patches) across 19 tissue types with 189,744 annotated nuclei in 5 cell categories. The dataset is pre-split into three folds for cross-validation. It is downloaded and verified; no dataset construction is required.

4 Plan of Activities

Weeks	Milestone
1–3	Data pipeline, baseline MLP classifier, proposal submission
4–5	Base U-Net implementation and training (VGG + ResNet encoders)
6–7	Swin Transformer encoder integration; CNN vs. Transformer comparison
8–9	Remaining ablations (BN, loss, depth, initialization)
10–11	Inference pipeline, visualization, failure analysis
12–14	Final report, ablation write-up, presentation

This is a solo project. All components—implementation, experiments, analysis, and writing—will be completed by Danny Rollo.

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